Bicol University POLANGUI CAMPUS

Centro Occidental Polangui, Albay

IT112: WEB SYSTEMS and TECHNOLOGIE'S

2nd Semester, AY: 2025-2026

Phase 3: Final Group Project

BentaBoard

"From bawat benta to tunay na kita—track it right."

Proposed By:

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BSIT 2A

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Presented to:

Dr. Guillermo V. Red, Jr.

IT112 Professor

Introduction

After finishing the backend and database setup in Phase 2, our group, "**PeNiHAMiNo**," moved on to "**Phase 3: Frontend Development**". This part of the project is where we turn our technical plans, like the ERD and Use Case diagrams, into actual web pages that a user can see and click on.

The goal of **BentaBoard** is to help small business owners move from "**bawat benta**" (every sale) to "**tunay na kita**" (actual profit). To make this happen, we needed a user interface (UI) that is simple, clean, and easy to use, even for people who aren't good in technology. We used HTML5, CSS3, and Bootstrap 5 to build the site. We chose Bootstrap because it makes the website "responsive," meaning it will look good and work correctly whether the user is opening it on a laptop, a tablet, or a smartphone.

In this phase, we focused on making sure that every page we built has a purpose. We didn't just make it look pretty; we made sure that every input field and button corresponds to the data we planned in our database. This ensures that when we connect the frontend to the backend in the next phase, everything will fit together perfectly.

Objectives

<u>At the end of this activity, we were able to:</u>

1. Translate project design documents (ERD, use case, architecture) into responsive HTML/CSS/Bootstrap interfaces.
2. Create modular, consistent, and functional frontend components that match the system's data model.
3. Prepare and connect pages for later backend/API interaction (planned in Phase 3).
4. Utilize Bootstrap components for layout, structure, and responsiveness.
5. Collaboratively commit their work to GitHub following a structured frontend workflow.

Frontend Folder Structure

```
/frontend
├── index.html
├── register.html
├── login.html
├── dashboard.html
├── inventory.html
├── sales.html
├── expenses.html
├── purchases.html
├── analytics.html
├── reports.html
├── settings.html
├── weather.html
├── /css/
├── /js/
└── /img/
```

I. Overview of Frontend Setup

Our frontend system is developed using HTML, CSS, and Bootstrap 5 to provide a modern and responsive user experience. To maintain consistency with our Phase 2 backend organization, we followed a modular folder structure.

The frontend is organized to separate the structure (HTML), the styling (CSS), and the logic (JS). This ensures that the project is easy to maintain and that multiple team members can work on different pages without causing code conflicts.

The frontend folder is structured as follows:

/frontend – The root directory containing all main HTML pages (e.g., index.html, dashboard.html).

/css – Contains custom CSS files to handle the specific branding and colors of BentaBoard.

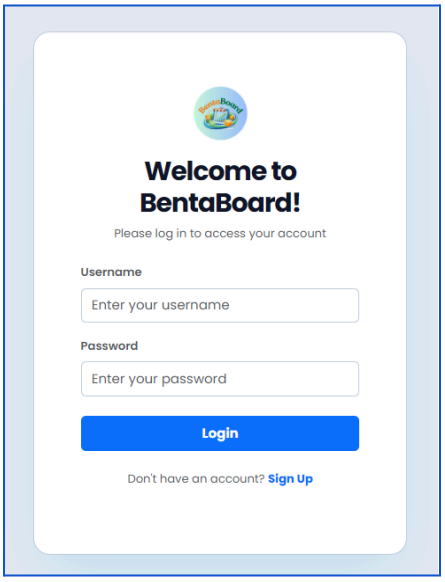
/js – Contains JavaScript files used for form validation and the future API integration scripts.

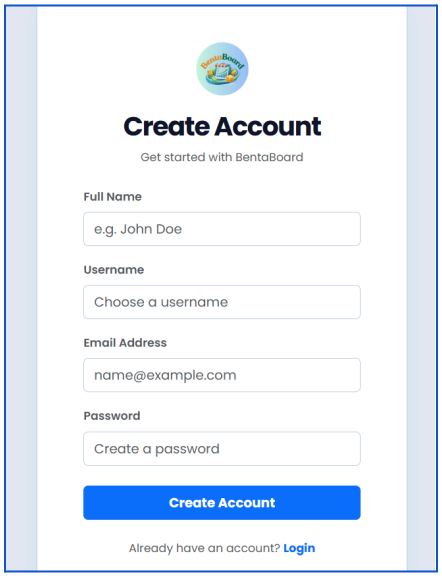
/img – Stores all optimized icons, logos, and images used across the interface.

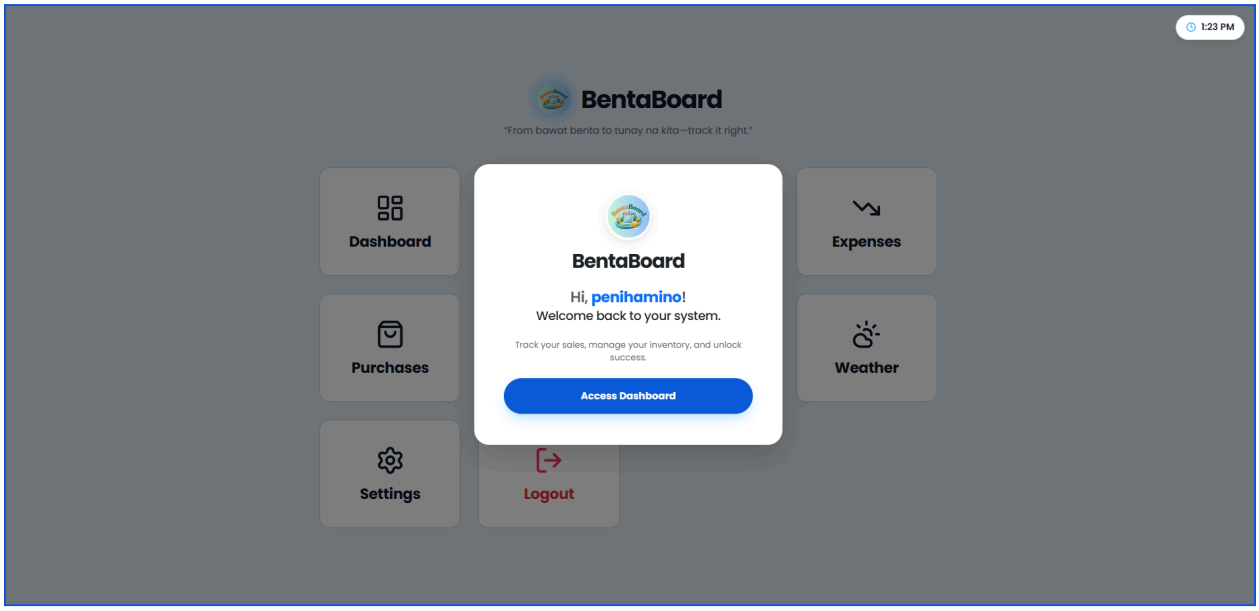
This setup ensures that the frontend is scalable, making it easy to add new pages or features as the project grows.

II. Screenshots of ALL working frontend pages

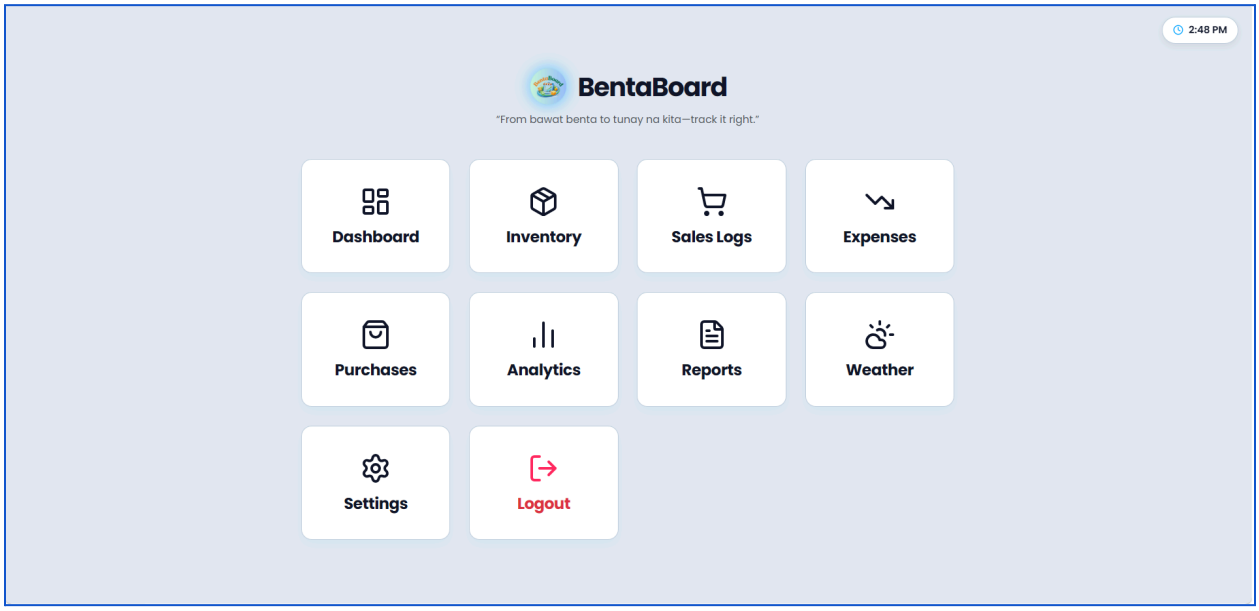
- **<u>Create/Login & Register/Sign Up</u>**



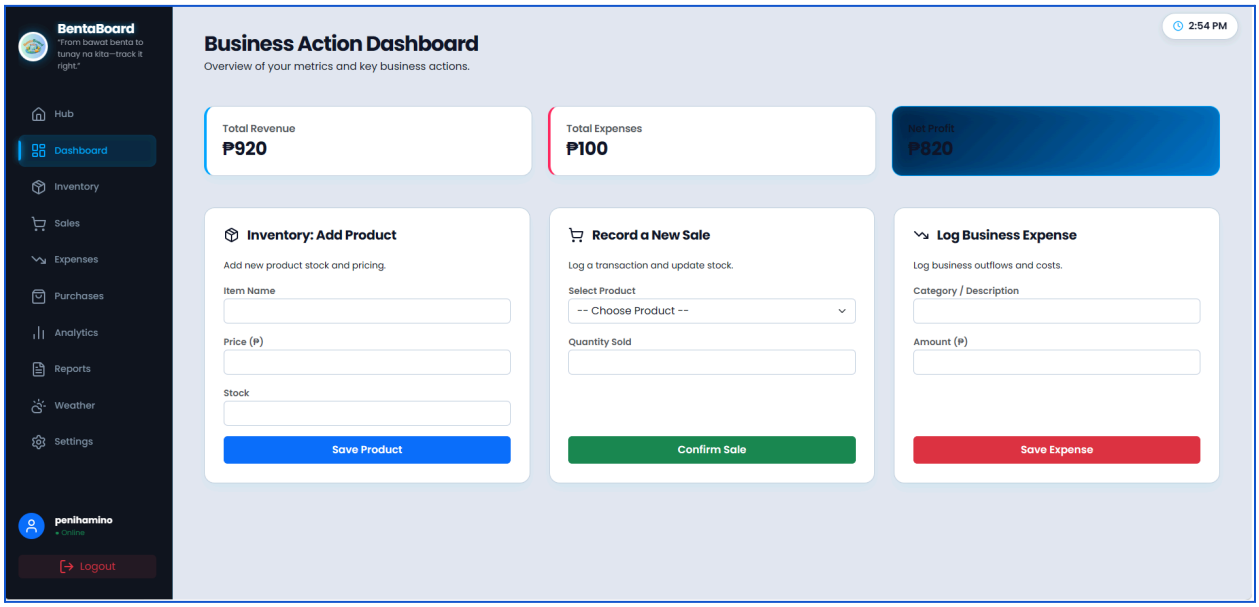



1:23 PM
BentaBoard
"From bawat benta to tunay na kita—track it right."
Dashboard
Expenses
Purchases
Weather
Settings
Logout
BentaBoard
Hi, penihamino!
Welcome back to your system.
Track your sales, manage your inventory, and unlock success.
Access Dashboard

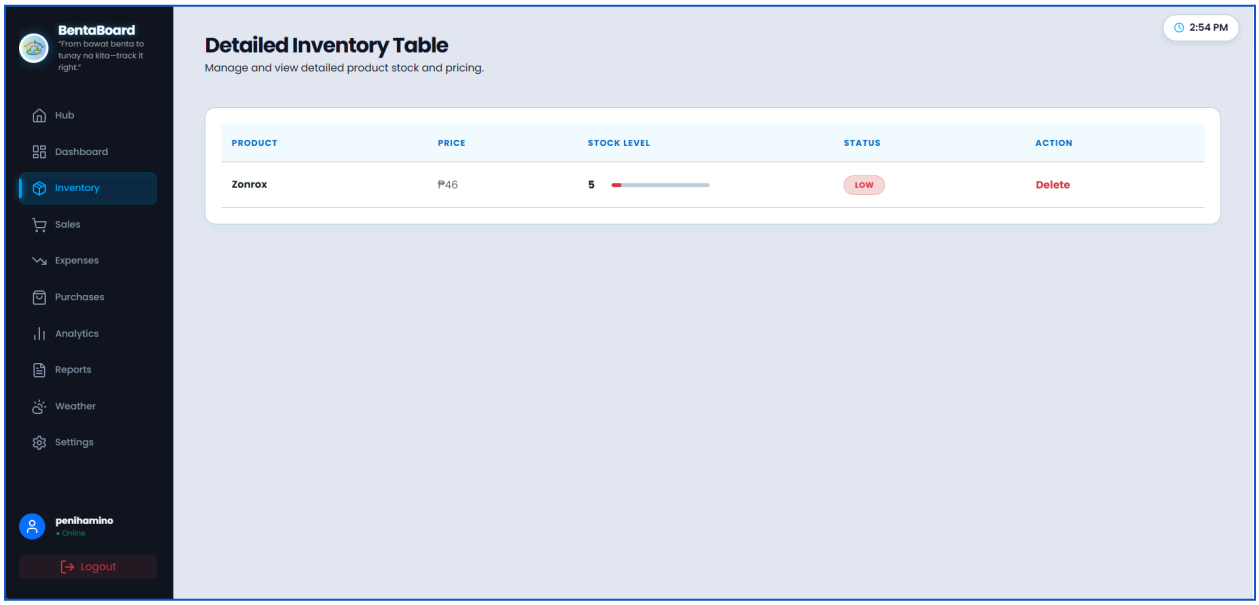
- **Hub**


2:48 PM
BentaBoard
"From bawat benta to tunay na kita—track it right."
Dashboard
Inventory
Sales Logs
Expenses
Purchases
Analytics
Reports
Weather
Settings
Logout

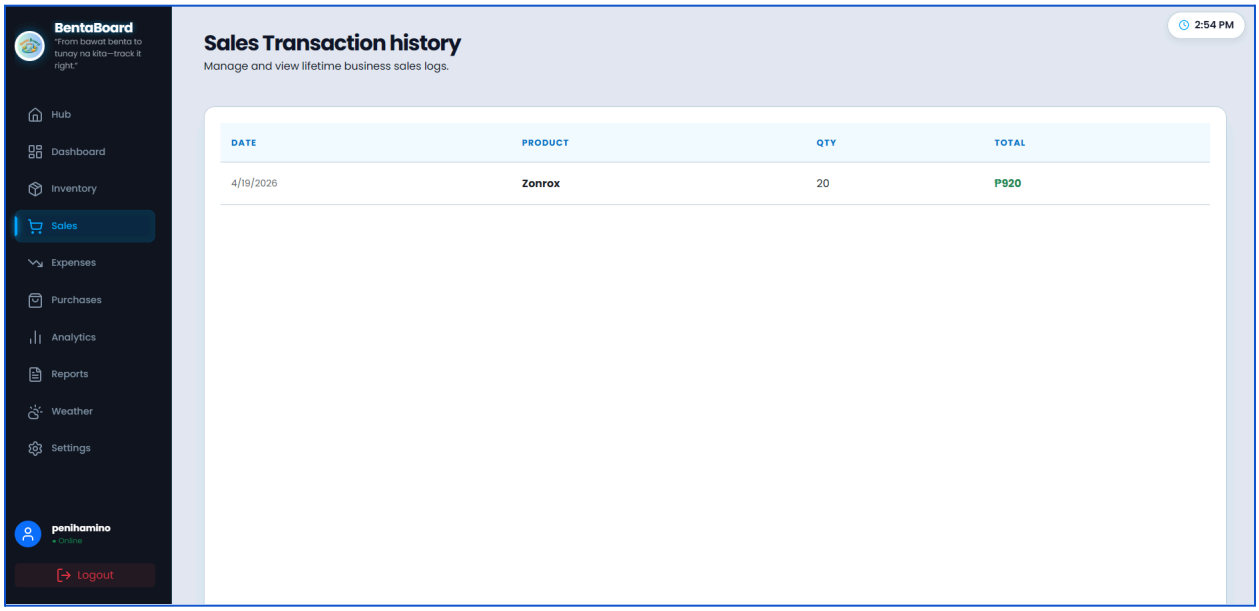
- **Dashboard**



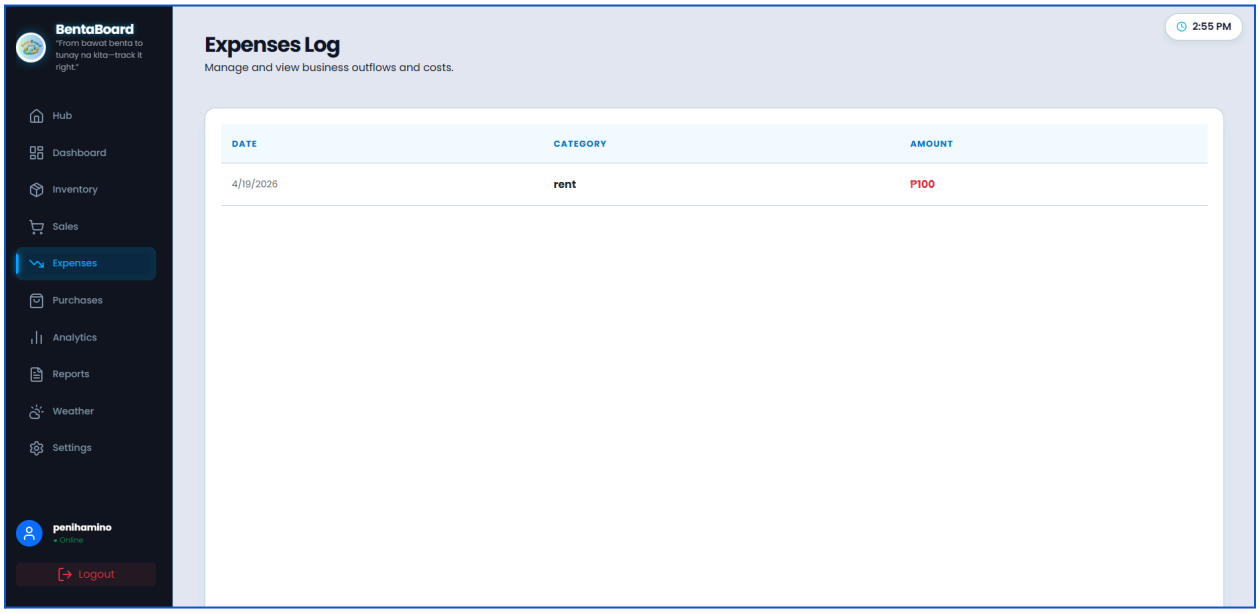
- **Inventory**



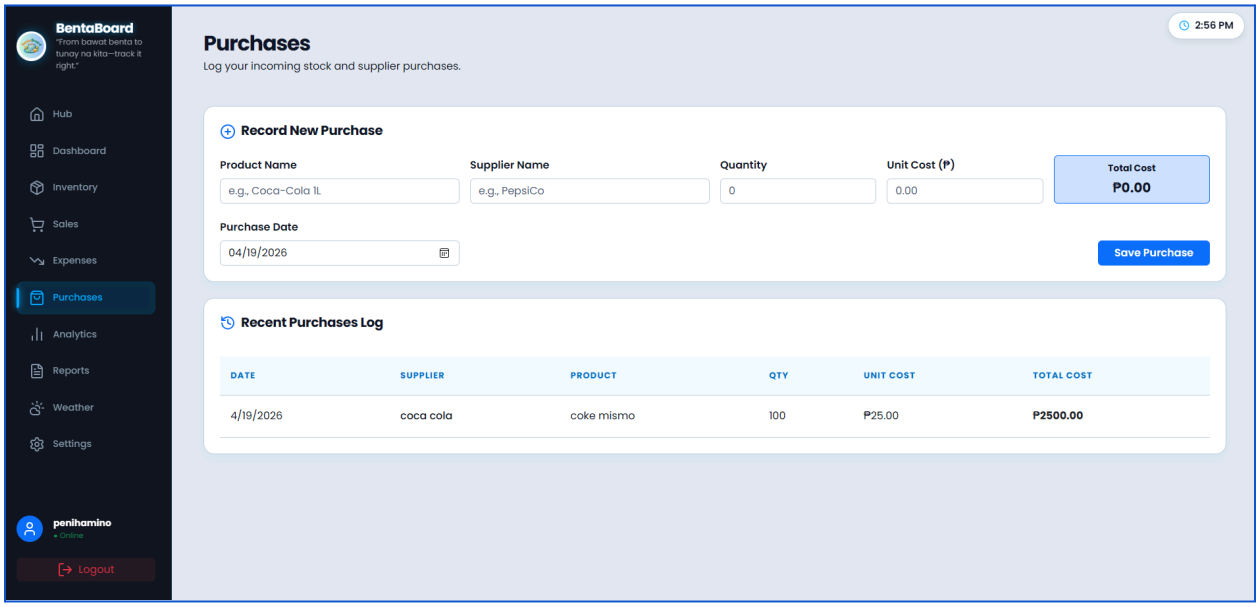
- **Sales**



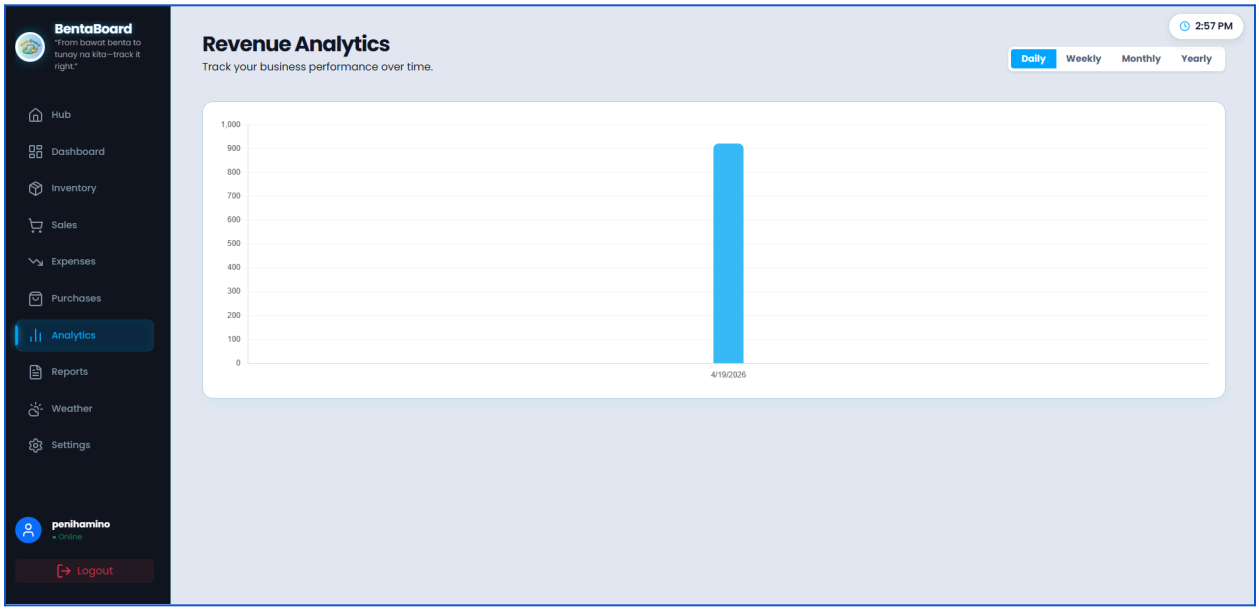
- **Expenses**



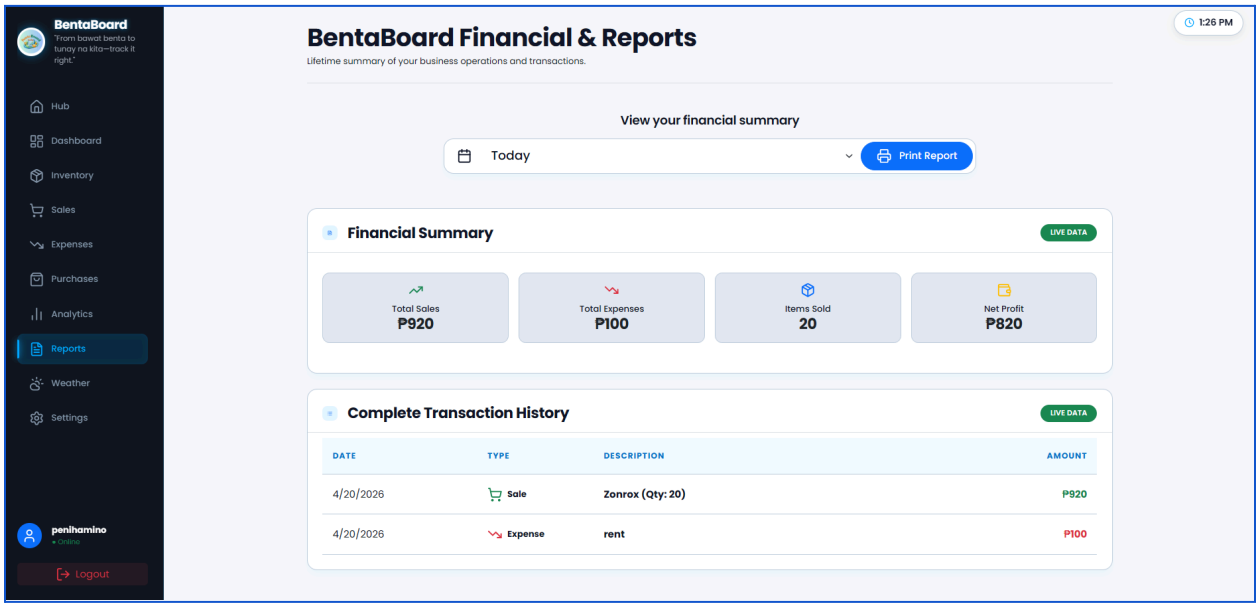
- **Purchases**



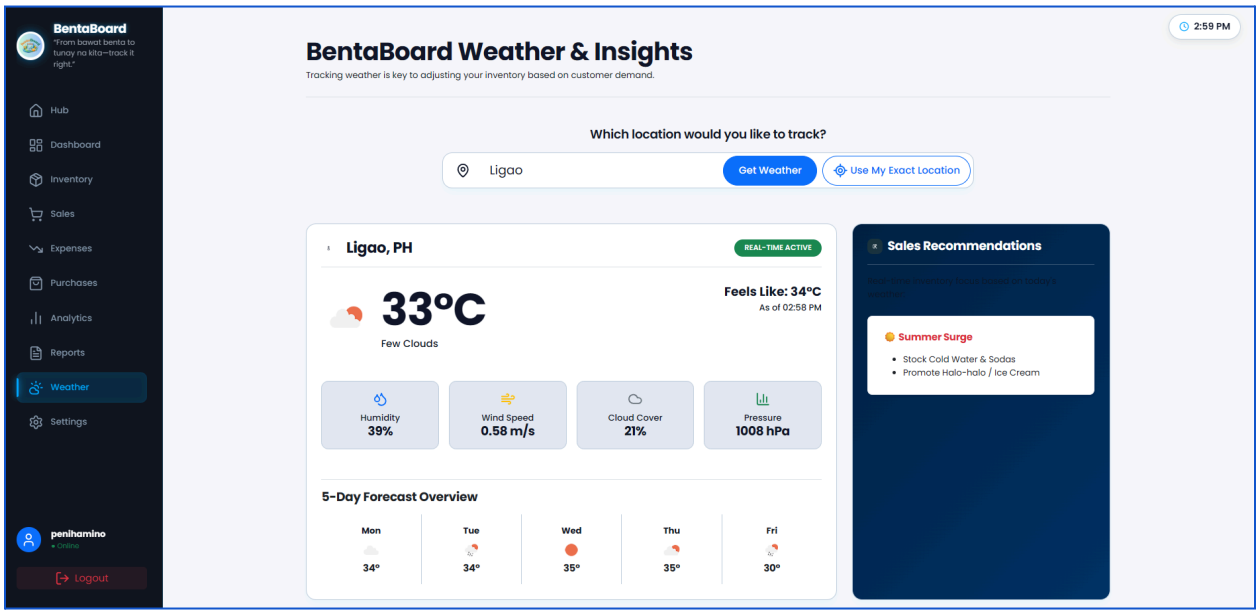
- **Analytics**



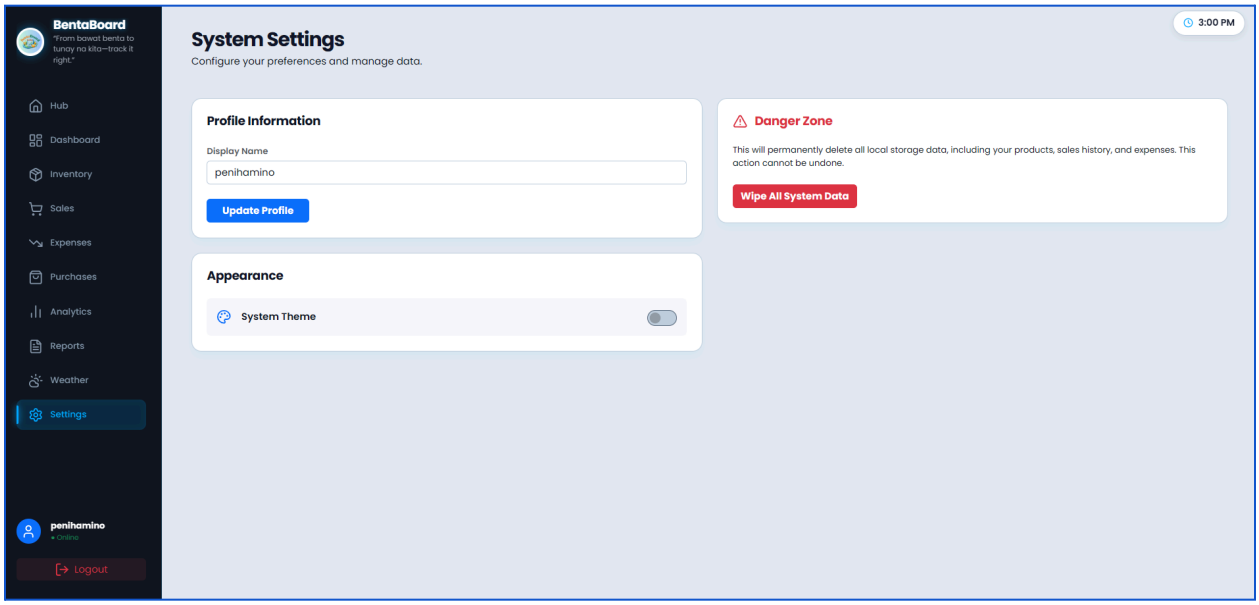
- **Reports**



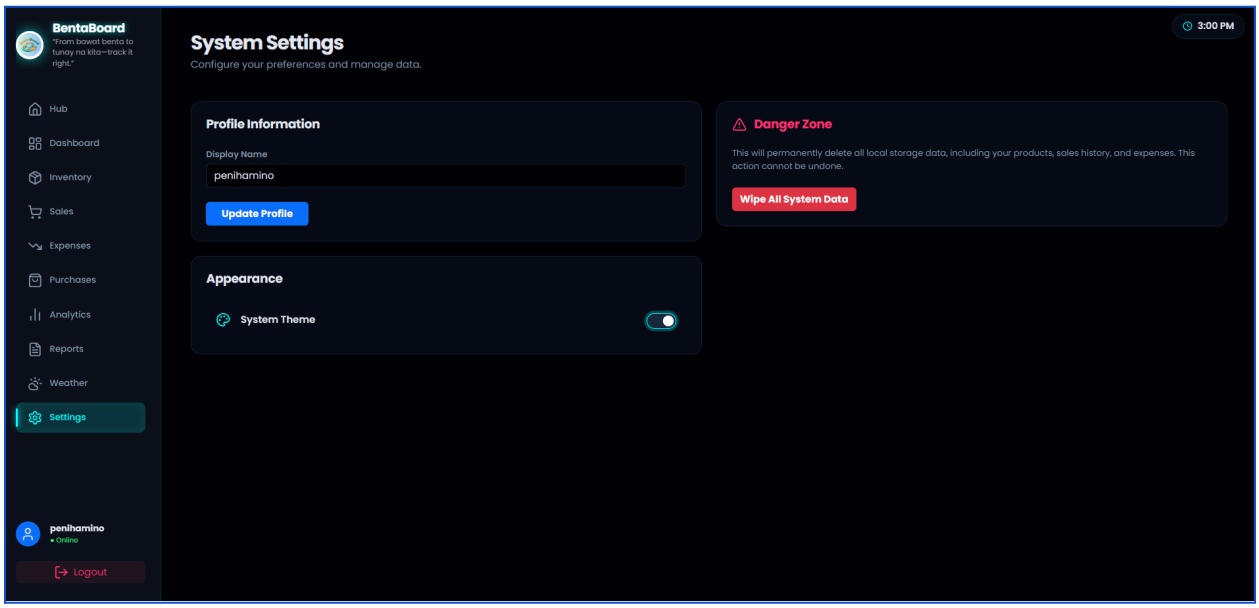
- **Weather**



- **Settings**



Darkmode

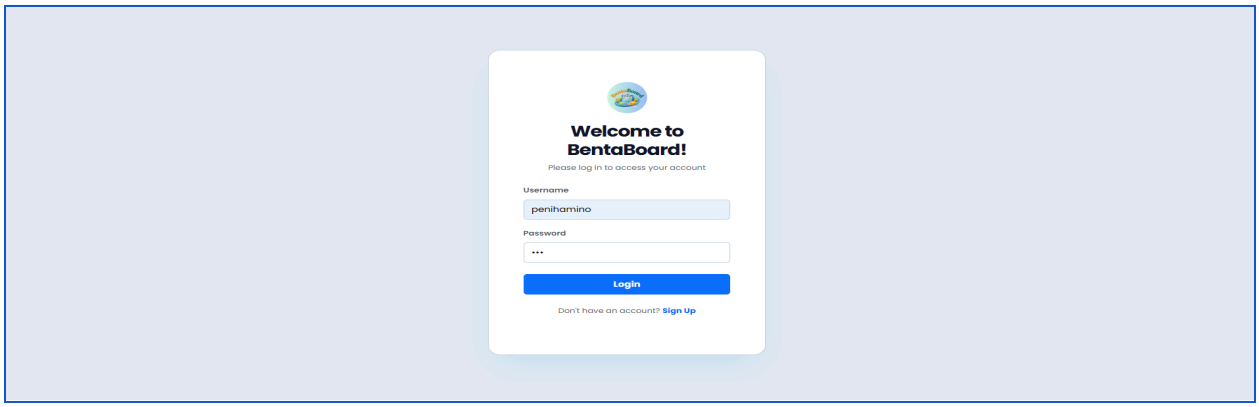


III. Annotation showing which feature/use case each page covers

We carefully followed our initial project proposal to ensure every feature we promised was included in the design. We organized our files into a /frontend folder to keep the project neat.

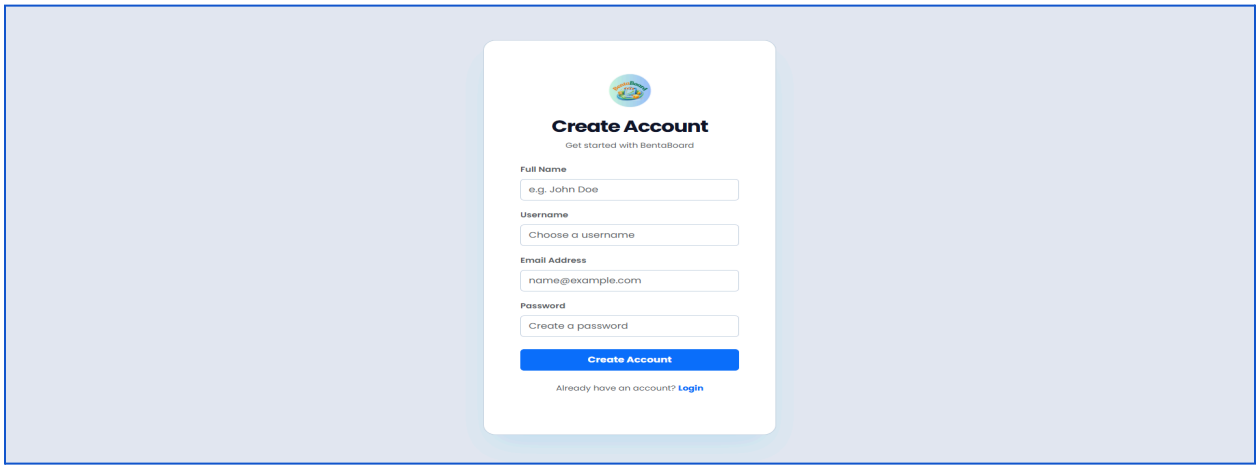
1. User Authentication (Login & Registration)

Based on our Use Case diagram, every user must be authenticated. We created a login and registration page with fields that match our User schema in the database.



- This page allows users to create an account and log in securely to access their business dashboard.

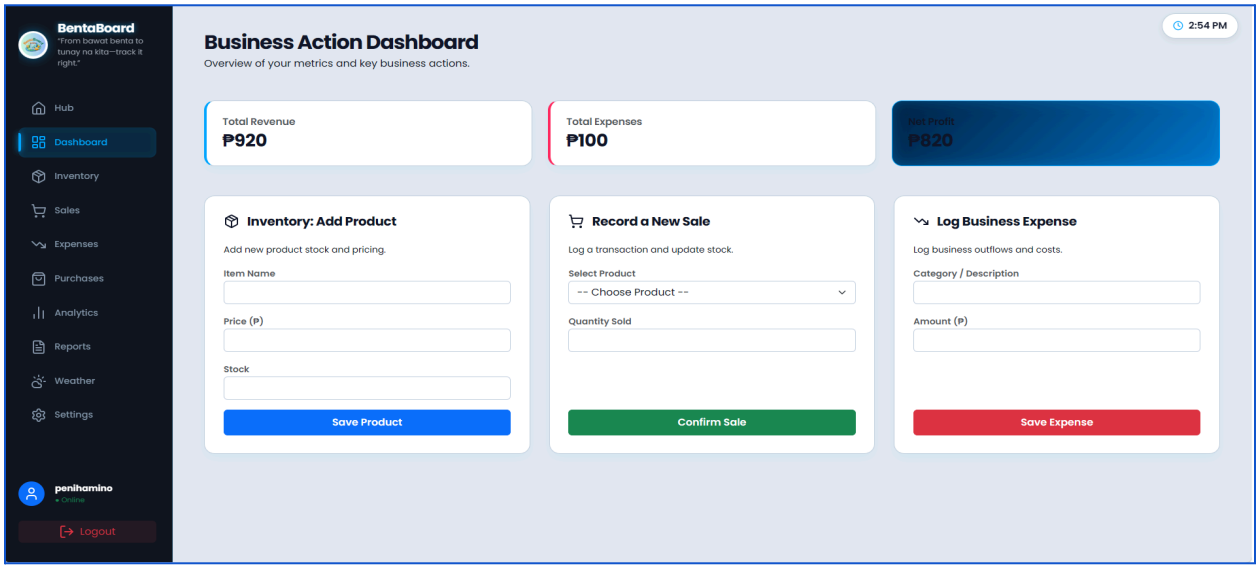
2. User Authentication (Sign Up Form)



- The sign-up form serves as the entry point to BentaBoard. It collects new user details like name, username, email, and password to create an account.

3. Main Dashboard

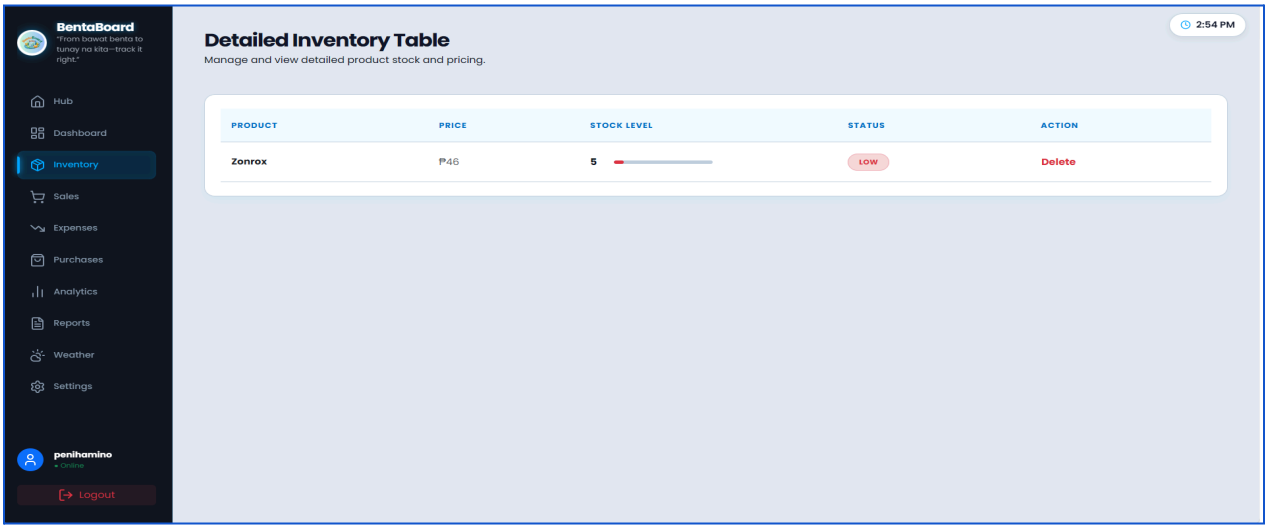
The dashboard serves as the central hub. It provides a quick overview of the business's financial health, showing total sales and profits.



- The dashboard gives the seller a snapshot of their daily and monthly earnings.

4. Detailed Inventory

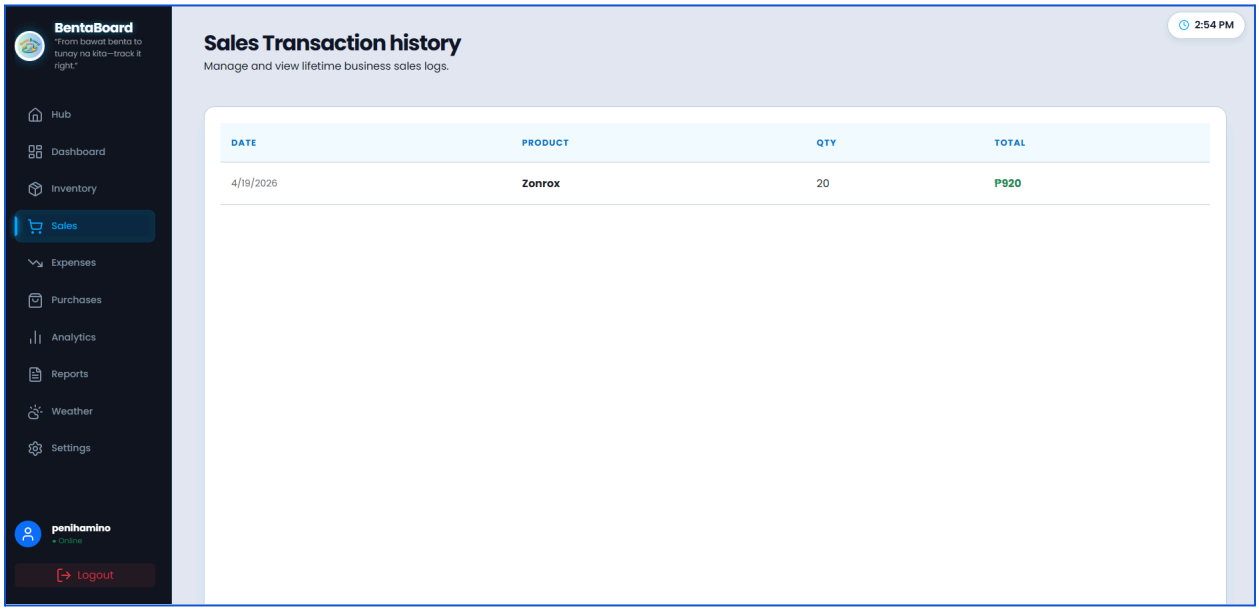
Following our ERD, we built a page where sellers can see the status of the products and delete their products. This page includes fields for product names, prices, stock levels, and status.



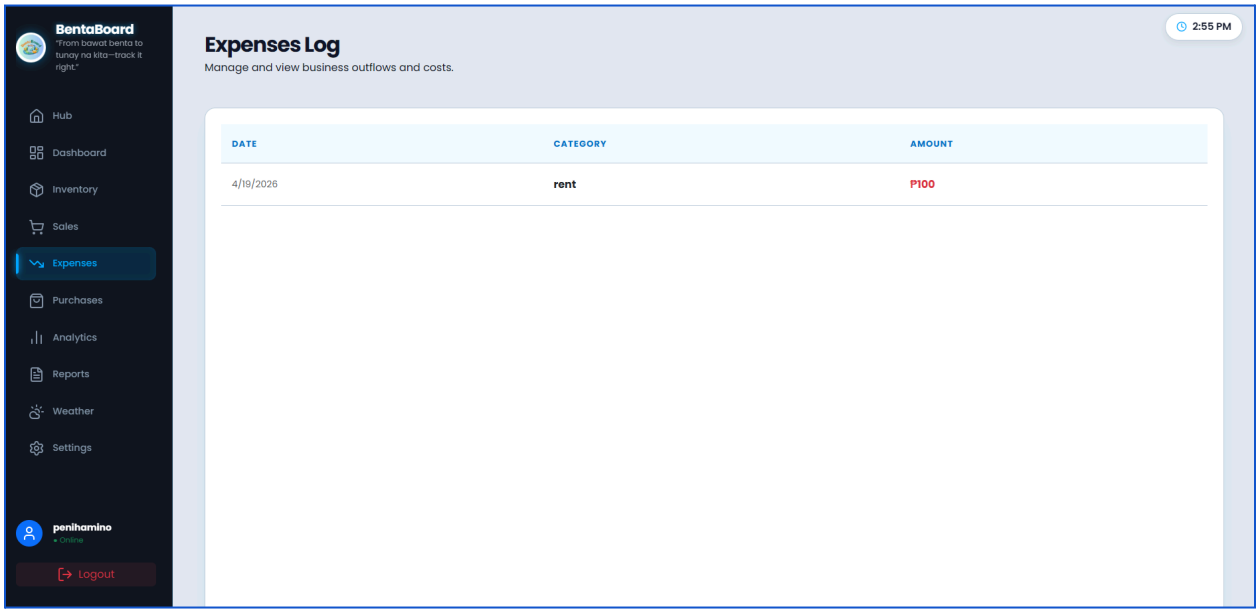
- This page handles the "Inventory Detailed" use case, allowing users to keep their stock list updated.

5. Sales and Expense Tracking

To fulfill the main purpose of BentaBoard, we created dedicated forms for recording daily sales and operational expenses.

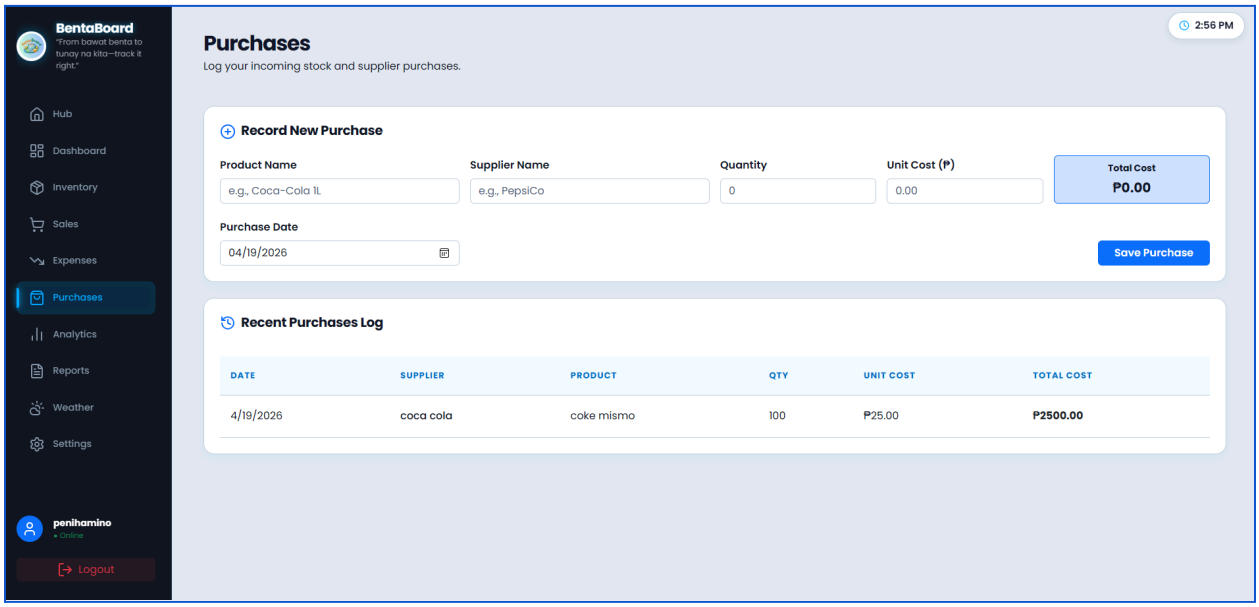


- This is where sellers log their daily transactions and quantities sold.



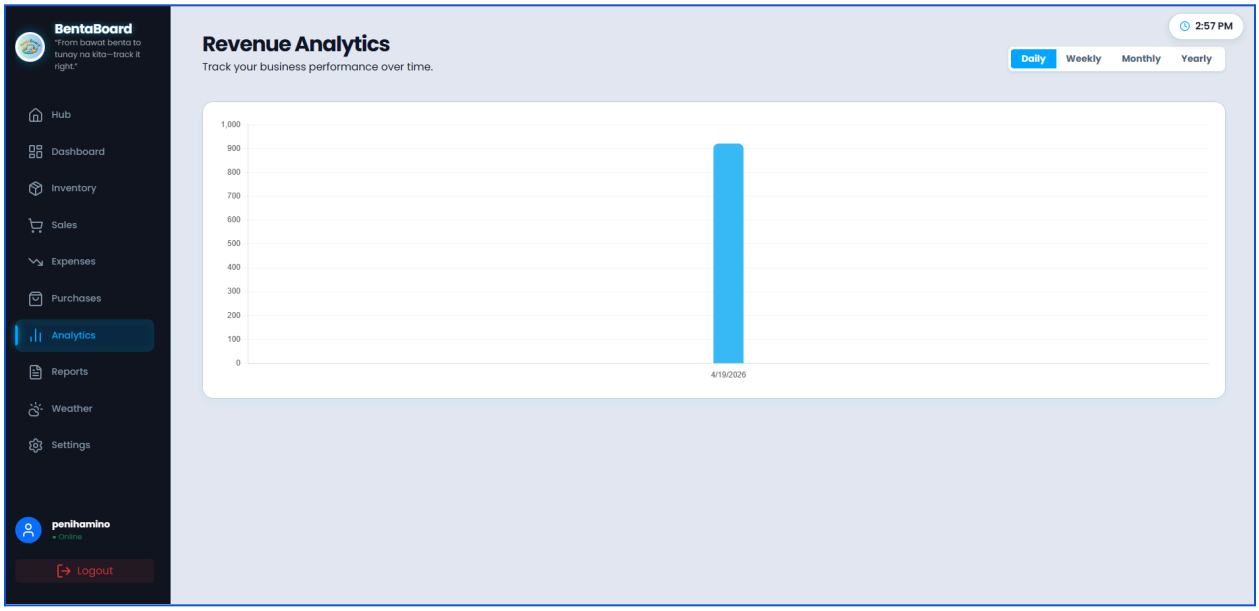
- This page allows users to track costs like rent, electricity, and transportation.

6. Purchases



7. Reports and Analytics

We designed a reports page that will eventually display the automated profit calculations (Total Sales - Expenses/Purchases).



- This page is designed to show the final profit reports on a daily, monthly, and yearly basis.

ISO 9001:2015
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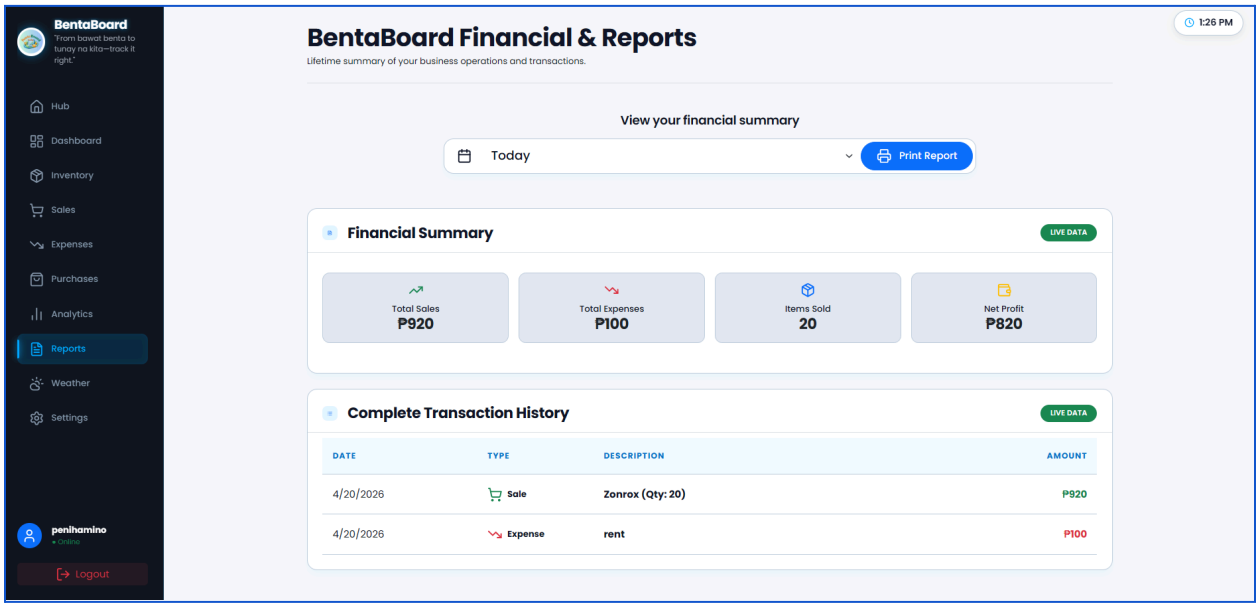
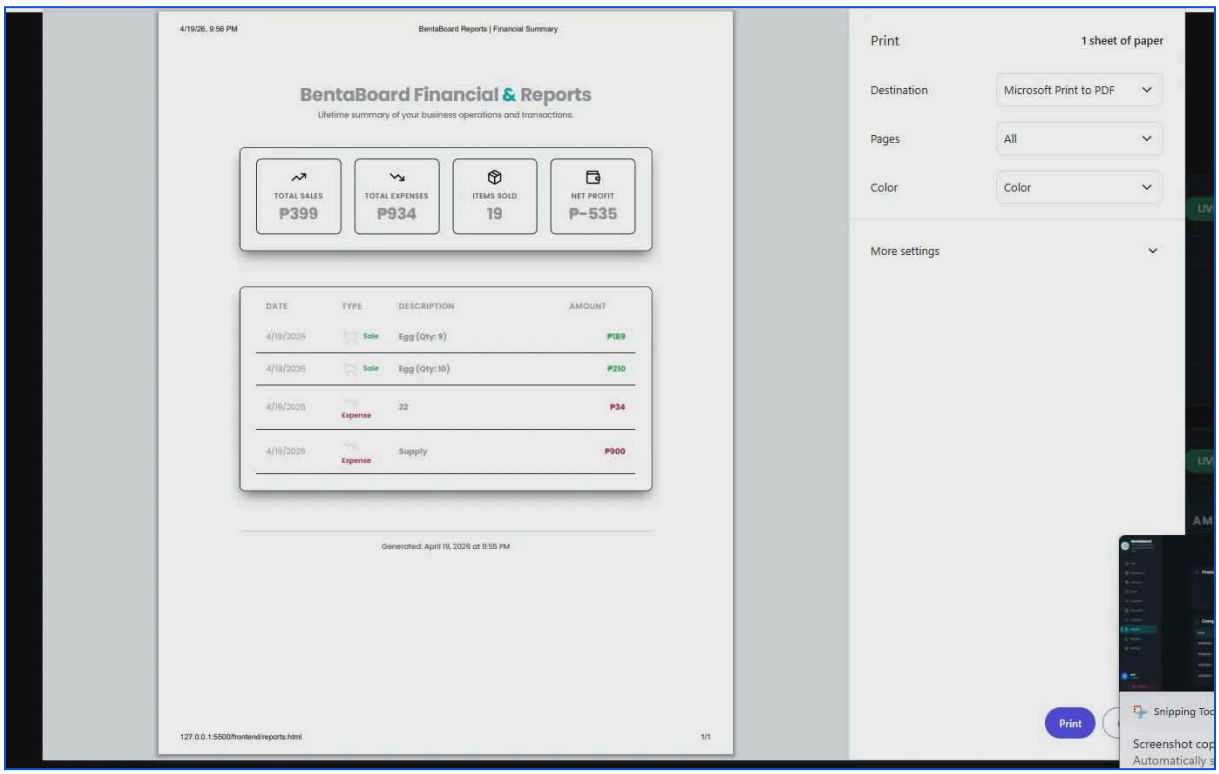
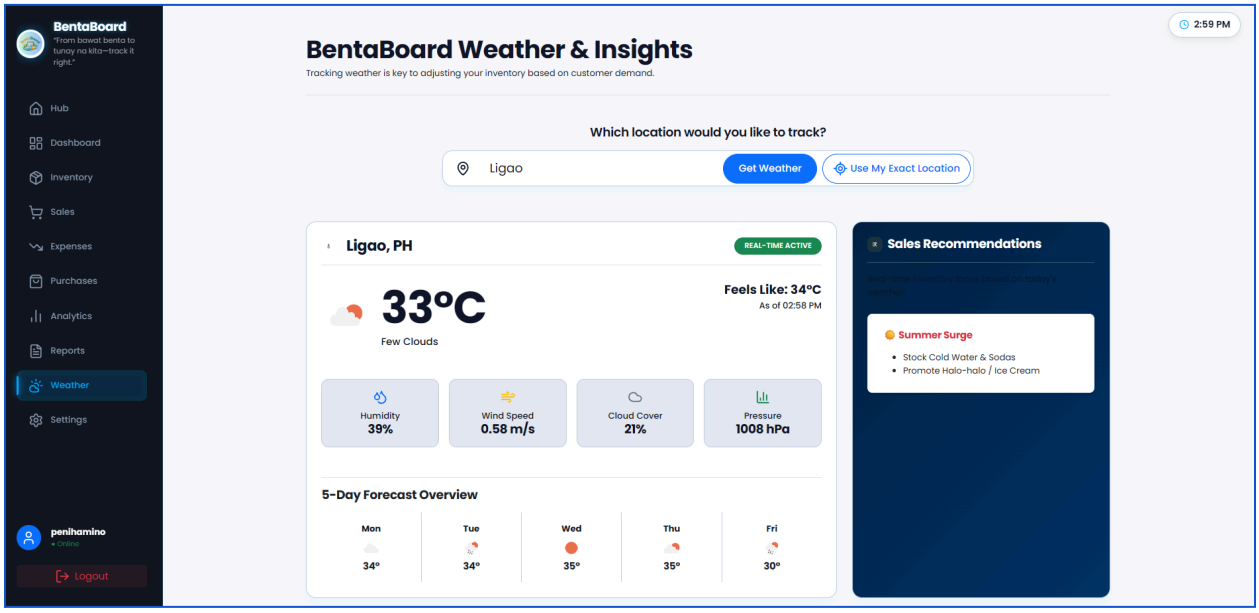
Report



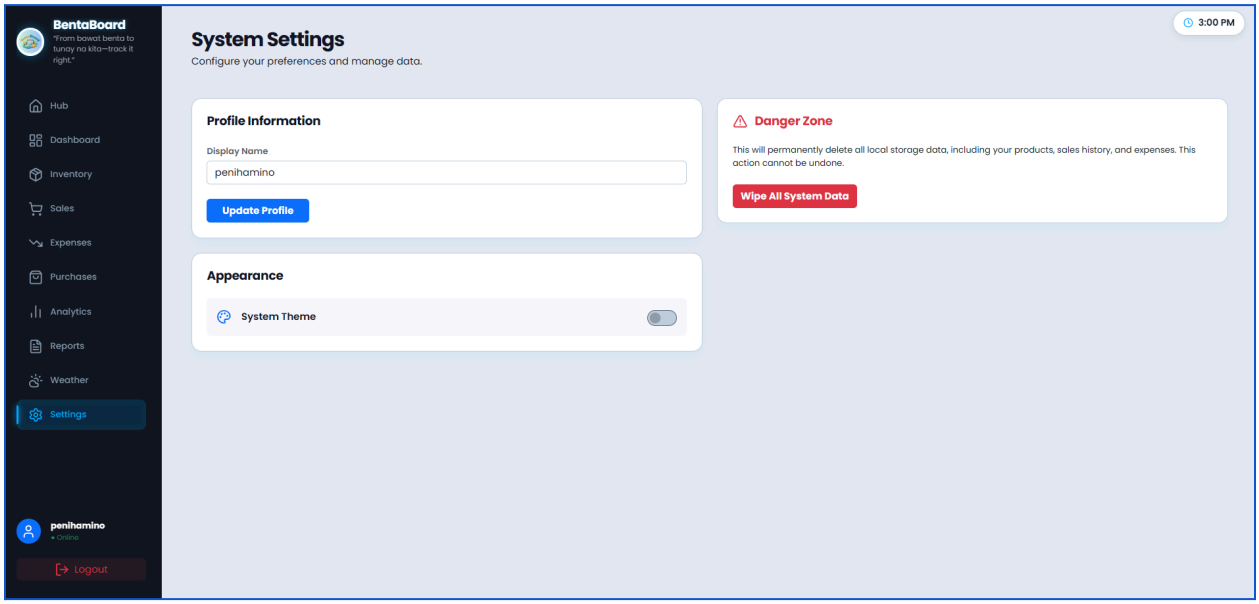


7. Weather



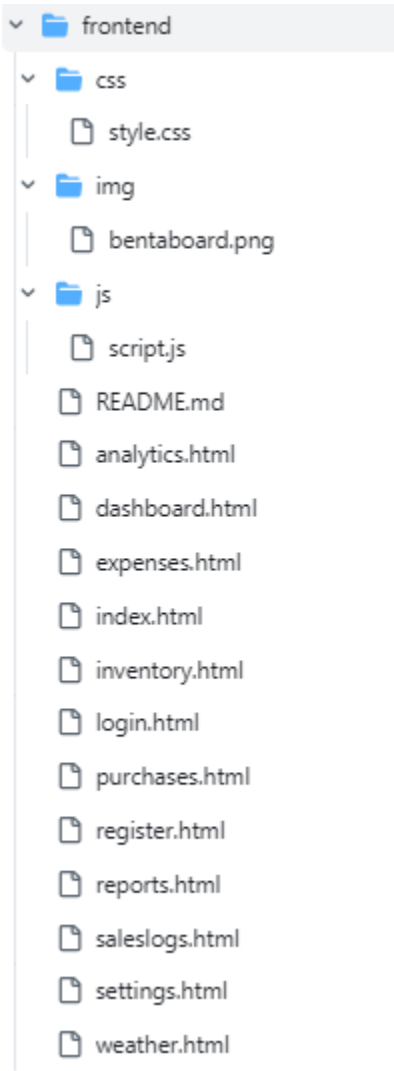
- We added a special page that shows the current weather. We did this because weather often affects how many customers visit a store. By having this integrated, the user can check the weather and plan their inventory or sales targets for the day.

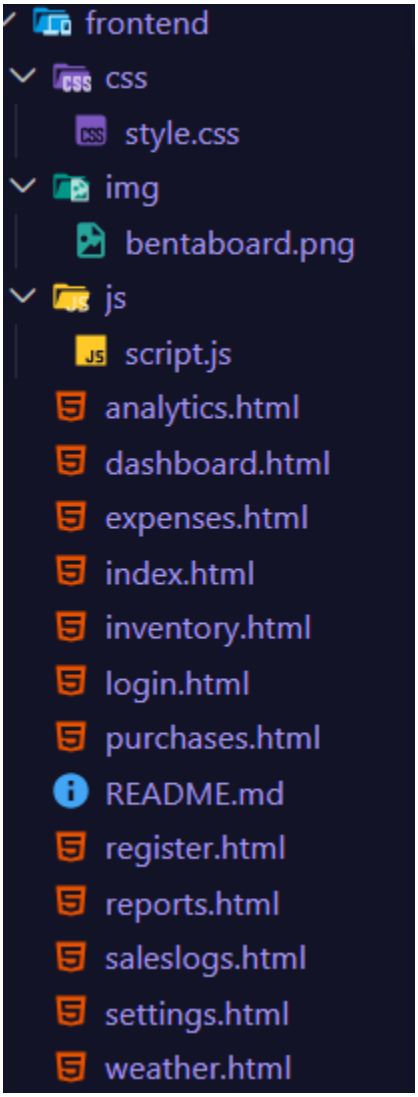
8. Settings



- We made a settings page. This is where the user can change their profile information or update their password. It's a simple page with a few input fields, making sure the user has full control over their account.

IV. Screenshot of GitHub folder structure





V. GitHub Commits

Commits on Apr 22, 2026

Cleanup: Remove .gitkeep from frontend folder prlanne committed 8 minutes ago	950ed2e
Added README.md in the frontend folder prlanne committed 14 minutes ago	3f71f0a

Commits on Apr 21, 2026

Added the frontend screenshots to the lab11 folder in docs directory prlanne committed 16 hours ago	cf22df3
Added script.js to frontend/js folder prlanne committed 16 hours ago	affbbce
Added bentaboard logo to frontend/img folder prlanne committed 16 hours ago	aa09994
Added style.css to frontend/css folder prlanne committed 16 hours ago	7b9a811
Added weather.html to frontend folder prlanne committed 16 hours ago	b3779ed
Added settings.html to frontend folder prlanne committed 16 hours ago	fd153f4
Added saleslog.html to frontend folder prlanne committed 16 hours ago	b42d1b5
Added reports.html to frontend folder prlanne committed 16 hours ago	239a022
Added register.html to frontend folder prlanne committed 16 hours ago	7c1728e
Added purchases.html to frontend folder prlanne committed 16 hours ago	5da01b0
Added login.html to frontend folder prlanne committed 16 hours ago	6958bbc
Added inventory.html to frontend folder prlanne committed 16 hours ago	49a308d
Added index.html to frontend folder prlanne committed 16 hours ago	6aefbd9
Added expenses.html to frontend folder prlanne committed 16 hours ago	752fff9
Added dashboard.html to frontend folder prlanne committed 16 hours ago	52aad1d
Added analytics.html to frontend folder prlanne committed 16 hours ago	77ece47

VI. Individual reflections from each group member (3–5 sentences)

Hazel Ann Obidece - Project Manager

As the Project Manager, my primary responsibility during this phase was ensuring that every developed interface strictly aligned with our original project proposal and technical goals. I focused on maintaining the scope of the project, carefully verifying that no promised feature from our initial design was overlooked during the coding process. While ensuring a natural and intuitive user flow across the website was challenging, I am proud that our team remained consistent with our vision.

Noe Jr Remolar - Documentation Officer

I spent my time taking screenshots of our progress and explaining how each page of our website works based on our original plans. I made sure that every button and form on the screen, like the 'Add Product' and 'Record Sale' sections, matches the diagrams we drew at the start of the project. It was very satisfying to see our ideas turn from simple drawings into a real, working website. This task taught me that having a clear and organized plan is the most important part of building a project because it makes the actual coding much easier and more accurate.

Michael Mantes - Database Manager

As the Database Manager, my primary focus was ensuring that every input field in our HTML forms aligned perfectly with the data types established in our MongoDB schemas. I meticulously verified each form field to prevent data mismatches that could compromise the system's functionality during full-stack integration. This process allowed me to deeply understand the critical connection between user interface design and back-end data management.

Nica Alyssa Racadag - FrontEnd Manager

Serving as the Frontend Manager, I led the actual development of our system's pages using the Bootstrap 5 framework to ensure a high-quality user experience. My main goal was to implement a responsive design that allows small business owners to manage their stores effectively on any device, especially mobile phones. I enjoyed experimenting with Bootstrap's card and navbar components to achieve a modern, high-tech aesthetic that matches our project's vision.

Pear Anne Odo - GitHub Manager

As the GitHub Manager, I was responsible for maintaining the integrity of our repository by coordinating team commits and organizing the frontend directory. I worked closely with each member to manage code pushes, successfully preventing conflicts and ensuring a smooth collaborative workflow. Additionally, I maintained the project's README file to provide clear instructions on how to navigate and interact with our newly developed pages. This role highlighted the importance of version control and organized documentation in managing a successful team-based technical project.